

9-2d Double-Declining-Balance Method

The **double-declining-balance method** (A method of depreciation that provides for a declining periodic depreciation expense over the expected useful life of an asset.) provides for a declining periodic expense over the expected useful life of the asset. The double-declining-balance method is applied in the following three steps:

Step 1.

Determine the straight-line percentage, using the expected useful life.

Step 2.

Determine the double-declining-balance rate by multiplying the straight-line rate (from **Step 1**) by 2.

Step 3.

Compute the depreciation expense by multiplying the double-declining-balance rate (from **Step 2**) times the book value of the asset.

To illustrate, the purchase of **Exeter**'s forklift is used to compute double-declining-balance depreciation. For the first year, the depreciation is \$9,600, computed as follows:

Step 1.

Straight-line percentage = 20% (100% ÷ 5)

Step 2.

Double-declining-balance rate = 40% (20% × 2)

Step 3.

Depreciation expense = \$9,600 (\$24,000 × 40%)

Depreciation of the forklift for the first year using the double-declining-balance method is recorded as follows:

Dec. 31	Depreciation Expense—Forklift	9,600	
	Accumulated Depreciation—Forklift		9,600

Assets	=	Liabilities	+	Stockholders' Equity
Accumulated Depreciation				Depreciation Expense
9,600				9,600

At the beginning of the first year, the book value of the equipment is its initial cost of \$24,000. After the first year, the book value declines, and thus, the depreciation also declines. The double-declining-balance depreciation for the full five-year life of the forklift is as follows:

Year	Cost	Acc. Dep. at Beginning of Year	Book Value at Beginning of Year		Double- Declining- Balance Rate	Depreciation for Year	Book Value at End of Year
1	\$24,000		\$24,000.00	×	40%	\$9,600.00	\$14,400.00
2	24,000	\$ 9,600.00	14,400.00	×	40%	5,760.00	8,640.00
3	24,000	15,360.00	8,640.00	×	40%	3,456.00	5,184.00
4	24,000	18,816.00	5,184.00	×	40%	2,073.60	3,110.40
5	24,000	20,889.60	3,110.40	—		1,110.40	2,000.00

When the double-declining-balance method is used, the estimated residual value is *not* considered. However, the asset should not be depreciated below its estimated residual value. In the preceding example, the estimated residual value was \$2,000. Therefore, the depreciation for the fifth year is \$1,110.40 (\$3,110.40 – \$2,000.00) instead of \$1,244.16 (40% × \$3,110.40).

Exhibit 7 illustrates the depreciation expense and book value of the forklift over its five-year life using the double-declining-balance method. As shown in **Exhibit 7**, the double-declining-balance method has higher depreciation in the first year of the asset's life, followed by declining depreciation amounts. For this reason, the double-declining-balance method is called an

accelerated depreciation method (A depreciation method that provides for a higher depreciation amount in the first year of the asset's use, followed by a gradually declining amount of depreciation.)

Exhibit 7

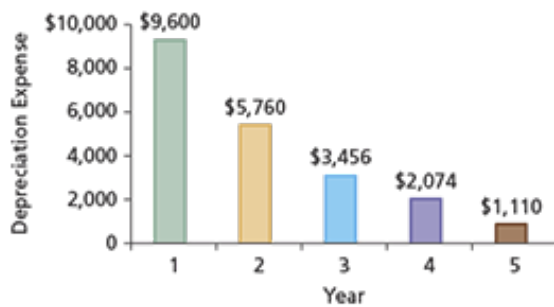
Double-Declining-Balance Method



Initial Cost.....	\$24,000
Residual Value.....	(2,000)
Depreciable Cost.....	<u>\$22,000</u>

	Year 1	Year 2	Year 3	Year 4	Year 5
Depreciation Expense	\$9,600.00	\$5,760.00	\$3,456.00	\$2,073.60	\$1,110.40 ← (\$3,110.40 – \$2,000.00)
Double-Declining-Balance Rate	40%	40%	40%	40%	40%

Depreciation Expense per Year



Book Value at Year-End



Revenues generated by an asset are often greater in the early years of its use than in later years. In such cases, the double-declining-balance method provides a good matching of depreciation expense with the asset's revenues.